

Digit Test Process

The Indian processor market is not completely in sync with the international market. As a result, you still find some 'phased out' processors in the market.

This throws up an interesting scenario where older processors may even compete against faster processors in the same price/performance bracket. An example of this is the wide availability of the older Coppermine series.

Keeping this in mind, we classified the processors by their positioning, based on the current price and the performance.



"Pairing today's processors with yesterday's technology is criminal!"

Hatim Kantawalla,
Deputy Head, Test Centre, Jasubhai Digital Media

Elite: This category includes the flagship processors of each manufacturer—the Pentium 4s and the Athlon XP.

Performance: This category exists purely because of the transitional period that the Indian processor market is currently undergoing. You'll find the Athlons and the Pentium IIIs battling it out here.

Value: Here you will see the scaled down versions of the

flagship models—the Durons and the Celerons.

Memory interface and other key issues: We used the fastest available memory solution, officially endorsed by the manufacturer for each family of processors. Hence, all Pentium 4s were tested with the fastest available RDRAM, whereas the AMD series was tested with the fastest available DDR-SDRAM (PC2100). The Pentium III was

tested with ordinary PC-133 SDRAM since it's the fastest 'official' solution till date. For more information on the benchmarks used, see *Digit CD Mindware*.

the fastest processor is not necessarily the one with the highest clock speed.

In a real-world environment, it's very difficult to say that a given processor is absolutely the fastest. Even with the same hardware, one processor may outshine the other depending on the applications in use and how it uses the processor instructions or the operating system. That's why a synthetic test (synthetic as in not being real-world) such as SiSoft Sandra is used to give a theoretical indication of how well the processor performs. The CPU benchmark of the test gives the performance indicators using the Dhrystone and Whetstone benchmarks. Dhrystone gives a result in MIPS (millions of Instructions per second) whereas Whetstone gives us a measure of the Floating Point Unit (FPU) and the scores are given in terms of MFLOPS (Millions FLoating OPerations per Second).

So, theoretically which is the fastest

CPU? Well, that depends on your application because in the Dhrystone benchmark (*refer Score board*), the Athlon XP 1800+ churned out 4337 MIPS as against 3909 MIPS for the Pentium 4, 2 GHz. However, in the Whetstone benchmark, the Pentium 4, 2 GHz scored 2452 MFLOPS as against 2137 MFLOPS for the Athlon XP 1800+. This indicates that, in theory, the Pentium 4 has more FPU strength than its Athlon XP counterpart and that, in theory, it will be faster when it comes to applications that are intensive in 3D rendering and image processing applications such as Photoshop.

Work on applications faster

You know what's theoretically faster, but these MIPS and MFLOPS scores from SiSoft Sandra are only indicative of the kind of performance that you could expect from each of these processors. If you want to know which processors will

make your daily use applications such as Word, Excel, Web authoring tools, etc faster, then the Business WinStone 2001 benchmark and the Content Creation WinStone 2001 test will tell you about it. Both these benchmarks run a series of applications such as an anti-virus suite, WinZip, Excel, Word, Adobe Premiere 6.0, Macromedia Director 8.5, and Macromedia Dreamweaver UltraDev 4. All of these are run in a scripted sequence and the total time consumed in performing each task is measured. The final score is a unit-less performance index.

An index of 40 for both these tests should be considered a bare minimum and a score below this should be considered as seriously low.

At the end of these tests, you will have the answer to the critical question: "How fast does this processor run 'my' application?"

The point to keep in mind here is that even if a certain processor takes only a

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The first microprocessor (Intel 4004) made for Busicom, a Japanese calculator manufacturer.

1974

Intel 80286 (the 286) with clock speeds of 8 MHz, 10 MHz, and 12 MHz, handled software applications written for 8088

1985

Intel 80486: This had the first built-in math coprocessor. It could handle GUI. AMD and Cyrix came up with products.

1993

First Generation
1971 Intel 8088 microprocessor (2 MHz clock speed; the brain of the first real personal computer, the Altair)

Second Generation
1982 Intel 386/386DX: Contained 250,000 transistors and was the first processor capable of handling 32-bit data lines. Speeds ranged from 16 to 33 MHz.

Third Generation
1989 Intel Pentium (P5/P54C): Intel changes the nomenclature to Pentium. Over 3 million transistors on the chip meant more high-tech programs, better Internet experience and more up time. AMD enters the fifth generation with K5, arriving a year late and at a lower clock speed than the Pentium.

Fourth Generation

Fifth Generation